

Pregnancy Outcomes in Patients After Completion of the mRNA Coronavirus Disease 2019 (COVID-19) Vaccination Series Compared With Unvaccinated Patients

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OBJECTIVE: To compare frequency of perinatal death between pregnant patients who completed the mRNA coronavirus disease 2019 (COVID-19) vaccination series and unvaccinated patients.

METHODS: This retrospective cohort study included 15,865 pregnant patients who delivered 16,132 newborns after 20 weeks of gestation within a large regional health system between January 1, 2021, and December 31, 2021. Patients who received two doses of mRNA vaccine (Pfizer–BioNTech [BNT162b2] or Moderna [mRNA-1273]) were included in the vaccinated group and were compared with unvaccinated patients. Exclusions included partial vaccination, viral-vector vaccine, major congenital anomalies, and higher-order multiple gestation. Our primary outcome was perinatal death, including stillbirth and neonatal death, which was evaluated by logistic regression. Unadjusted odds ratios and adjusted odds ratios (aORs) were reported, controlling for age, body mass index (BMI), diabetes, hypertension, smoking, twin gestation, and insurance status. Propensity score matching was also performed.

RESULTS: A total of 15,865 patients were included in the final analysis: 2,069 in the vaccination group and 13,796

in the control group. Only 13.0% of the cohort was included in the vaccination group; however, the vaccination rate increased over the course of the study period as the vaccine became more widely available and accepted. Vaccinated patients were older, with higher rates of people of non-Black racial non-Hispanic ethnic backgrounds, people with private insurance, and those with higher BMIs. Vaccination was associated with a lower incidence of perinatal death (0.5% vaccinated group vs 0.8% unvaccinated group, aOR 0.20 0.05–0.88). Vaccination against COVID-19 was also associated with lower rates of preterm delivery (aOR 0.63, 0.48–0.82), neonates with very low birth weight (aOR 0.35, 0.15–0.84), and neonatal intensive care unit (NICU) admission (aOR 0.66, 0.52–0.85). The association between vaccination and lower rates of perinatal death was no longer significant after propensity score matching.

CONCLUSION: In a large retrospective cohort study, receipt of the primary mRNA COVID-19 vaccination series was associated with a lower rate of several adverse pregnancy outcomes, including perinatal death, preterm delivery, neonates with very low birth weight, and NICU admission. Although the decreased rates of perinatal death did not remain significant after propensity score matching, there was evidence of directional benefit for vaccinated patients.

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Guidance from the Centers for Disease Control and Prevention (CDC), the Society for Maternal-Fetal Medicine, and the American College of Obstetricians and Gynecologists recommends coronavirus disease 2019 (COVID-19) vaccination for all



people 6 months or older, including pregnant and lactating individuals.^{1–3} Despite strong recommendations and clear maternal benefit for vaccinated pregnant patients,^{4,5} vaccine uptake among pregnant patients lags behind other high-risk groups with persistent regional variations.⁶

Vaccine hesitancy is a familiar phenomenon in pregnancy and is not unique to the COVID-19 vaccine.⁷ Hesitancy surrounding the COVID-19 vaccine in pregnancy is centered around preexisting vaccine hesitancy in the pregnant population, the spread of misinformation regarding the risks of the COVID-19 vaccine to pregnant and lactating individuals, the lack of data regarding safety due to exclusion of pregnant and lactating individuals from initial vaccine trials, and the perception that individual risk is low in young healthy reproductive-aged females.⁷ It is important for the medical community to continue to provide new data with regard to the safety and efficacy of COVID-19 vaccines in pregnancy.

Several studies have shown an association between severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection in pregnancy and increased rates of stillbirth, adverse neonatal outcomes, preterm birth, and low birth weight, particularly when maternal infection is severe.^{8–12} Several small retrospective studies and a recent population-based study aimed at assessing safety of COVID-19 vaccines in pregnancy have highlighted the finding that vaccination in pregnancy is not associated with an increased risk of adverse pregnancy outcomes.^{13–15}

Early in the Delta variant phase of the pandemic, we demonstrated that vaccination in pregnancy was associated with a lower risk of severe or critical maternal COVID-19.⁴ We hypothesize that vaccination against COVID-19 in pregnancy is also protective against adverse perinatal outcomes. The primary objective of this study is to evaluate whether COVID-19 vaccination during pregnancy is associated with a lower frequency of perinatal death.

METHODS

This study was approved by the Ochsner Health System IRB (IRB ID: 2020.142) (DHHS Federal Wide Assurance Identifier: FWA00002050). A full Health Insurance Portability and Accountability Act waiver of consent was granted given the use of a de-identified retrospective data set. Our study included all pregnant patients delivering after 20 weeks of gestation within the Ochsner Health System between January 1, 2021, and December 31, 2021. The majority of the study period occurred during SARS-CoV-2 Alpha (B.1.1.7) and Delta (B.1.617.1) variant

dominance.¹⁶ The less virulent Omicron (B.1.1.529) variant became the dominant strain of SARS-CoV-2 in late December 2021, the last 2 weeks of our study period.¹⁶ Viral particles were not routinely genotyped during the study period in our health care system.

Pregnant patients were identified for inclusion by using delivery contact serial numbers within the Ochsner Health Epic electronic medical record system. Contact serial numbers are unique identifiers within the medical record for individual patient encounters such as an admission, obstetrics delivery, or outpatient visit. Patients identified for inclusion were then divided into two groups, those who were received the full course of mRNA COVID-19 vaccine and those who were unvaccinated. To be considered vaccinated, patients must have received two doses of either the Pfizer-BioNTech (BNT162b2) or Moderna (mRNA-1273) vaccine before delivery. Patients with SARS-CoV-2 infection during pregnancy before receiving the first dose of the vaccine were included in the unvaccinated group. Patients who received only one dose of the vaccine were excluded, as were patients who received viral vector (Johnson & Johnson/Janssen) vaccines. Patients who received the viral vector vaccine were excluded due to recent safety concerns that prompted the CDC to recommend the viral vector vaccine only in certain rare clinical circumstances.¹⁷ Confirmation of vaccine status, as reported in the Epic medical record system, was improved by the integration of the Louisiana Immunization Network within the electronic medical record, allowing ascertainment of vaccination status regardless of where the patient received the vaccination in our health system. Vaccination status was also validated through manual medical record review to ensure no vaccinated patients were inadvertently included within the unvaccinated group. Medical record review was also performed to examine all stillbirths to exclude any stillbirths with known major congenital anomalies, as well as for preterm deliveries to determine whether preterm deliveries were spontaneous or medically indicated. The remainder of our data were extracted from the electronic medical record by our informatics personnel. Additional exclusions from final analysis included delivery before 20 weeks of gestation, higher-order multiple gestations, and major congenital anomalies (including known chromosomal abnormalities, hydrops, cardiac, or other anomalies associated with increased risk of the primary outcome).

Our primary outcome was perinatal death, including stillbirth and neonatal death. *Stillbirth* was defined as fetal death after 20 weeks of gestation or



fetal weight greater than 350 g. *Neonatal death* was defined as death in the first 28 days of life. Secondary neonatal outcomes included preterm delivery before 37 weeks of gestation, small for gestational age neonates (birth weight less than the 10th percentile on the Fenton¹⁸ curve by neonatal sex), very low birth weight (birth weight less than 1,500 g), and neonatal intensive care unit (NICU) admission. Maternal outcomes were also collected, including maternal SARS-CoV-2 infection during pregnancy, active SARS-CoV-2 infection at time of delivery, severe or critical maternal COVID-19, and preeclampsia. Institutional policy during the study period required all patients being admitted to the hospital to have a SARS-CoV-2 test. Severe or critical maternal COVID-19 was determined using the National Institutes of Health definition of the clinical spectrum of SARS-CoV-2 infection.¹⁹ *Severe illness* was defined as SpO₂ levels of less than 94%, a PaO₂/FiO₂ ratio less than 300 mm Hg, respiratory rate greater than 30 breaths per minute, or lung infiltrates greater than 50%.¹⁹ *Critical illness* was defined as patients with respiratory failure, septic shock, or multiple organ dysfunction.¹⁹

Demographic and baseline clinical characteristics potentially associated with vaccination status or adverse neonatal or maternal outcomes from COVID-19 were collected from the electronic medical record. Covariates included maternal age at delivery, nulliparous status, Black racial designation, Hispanic ethnicity, insurance type (Medicaid vs private), prepregnancy body mass index (BMI, calculated as weight in kilograms divided by height in meters squared), smoking status, maternal medical comorbidities (diabetes, hypertension, human immunodeficiency virus, asthma), and twin gestation.

Categorical variables were compared using χ^2 test and Fisher exact test, where appropriate. Means of continuous data were compared using the *t* test. For primary and secondary outcomes, odds ratios (ORs) and 95% CIs were generated using logistic regression. Analyses were adjusted for predictors of COVID-19 and perinatal outcomes, including maternal age, BMI, diabetes, hypertension, current smoking status, Medicaid insurance, and twin gestation. Because twin gestation is associated with a range of adverse outcomes, a sensitivity analysis that included only singleton gestations was performed with outcomes evaluated again by logistic regression. Finally, propensity score matching analysis²⁰ was performed in an attempt to balance the cohorts based on the likelihood of exposure (ie, vaccination), with matching performed using the *MatchIt* package in R.²¹ Because a large proportion of baseline BMI data were missing, a model limited

to only complete data was evaluated. Potential propensity score estimate models were generated using logistic regression, generalized additive models, artificial neural network, and classification and regression tree with full matching due to the infrequency of the primary outcome. The optimal model was determined based on balance diagnostics, with targets for standardized mean difference less than 0.10 and variance ratios between 0.5 and 2.0. Determination of average treatment effect on the treated was determined using logistic regression to facilitate double robustness.²² Analysis was performed using R.²³

RESULTS

A total of 16,467 pregnant patients were identified during the study period, with 602 pregnant patients excluded. The final study population included 15,865 pregnant patients, with 2,069 (13.0%) in the vaccination group and 13,796 (87.0%) in the unvaccinated group (Fig. 1). A total of 16,132 newborns were delivered when accounting for twin gestations. The percentage of vaccinated patients increased throughout the study period, with fewer than 0.01% (1/1,307) of patients who delivered in January 2021 having received two doses of vaccine before delivery compared with 30.8% (401/1,304) of patients who delivered in December 2021 having received two doses of vaccine before delivery.

Vaccinated patients were more likely to be older, of non-Black racial background, and of non-Hispanic ethnicity (Table 1). We also found an association among nulliparity, private insurance, and current non-smoking status and a higher rate of vaccination. Prevalence of twin gestation and medical comorbidities, which include preexisting diabetes, hypertension, human immunodeficiency virus (HIV) infection, and asthma, were similar between groups. Route of delivery was also similar between groups.

Vaccination was associated with a 44% reduction in perinatal death (0.5% vs 0.8%, adjusted odds ratio [aOR] 0.20, 0.05–0.88, Table 2). Examined individually, there were lower odds of stillbirth (0.4% vs 0.7%, aOR 0.13, 0.02–0.99), but not neonatal death (0.05% vs 0.17%, OR 0.27, 0.01–1.7) between the two groups. Additionally, the vaccinated group had lower odds of preterm delivery before 37 weeks of gestation (10.3% vs 13.2%, aOR 0.63, 0.48–0.82), indicated preterm delivery (5.6% vs 6.8%, aOR 0.60, 0.42–0.86), very low birth weight (1.1% vs 2.1%, aOR 0.35, 0.15–0.84), and NICU admission (11.8% vs 15.3%, aOR 0.66, 0.52–0.85, Table 2).

A sensitivity analysis was performed restricting the cohort to only singleton gestations. The



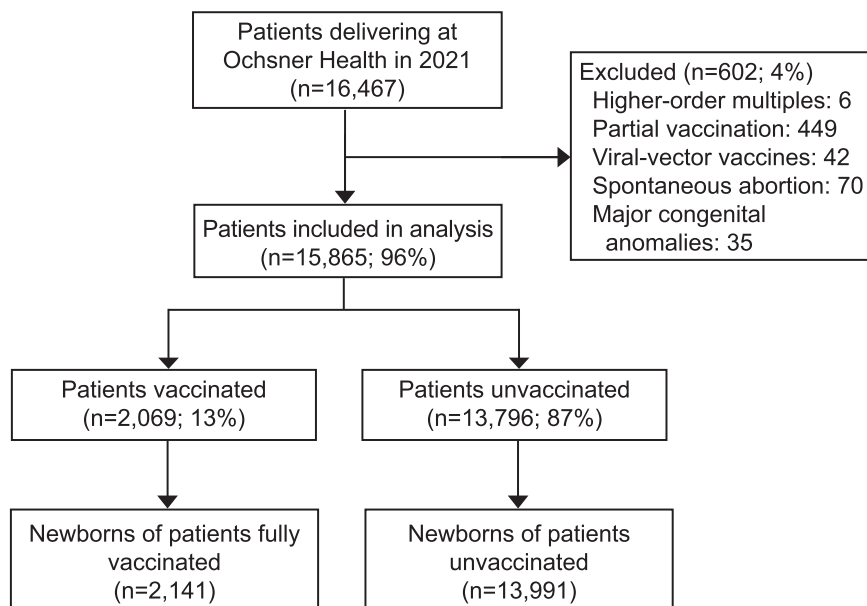


Fig. 1. Patient inclusion flow diagram.

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association between vaccination and lower incidence of perinatal death remained significant (0.4% vs 0.8%, aOR 0.1, 0.01–0.75). Individual outcomes of stillbirth and neonatal death were not significantly different between the two groups. Vaccination remained associated with lower rates of preterm delivery (9.2% vs 12.4%, aOR 0.63, 0.48–0.83), indicated preterm delivery (4.7% vs 6.4%, aOR 0.52, 0.35–0.77), NICU admission (10.4% vs 14.3%, aOR 0.66, 0.51–0.86), and very low birth weight (3.3% vs 3.7%, aOR 0.28, 0.1–0.79).

With regard to maternal outcomes, vaccination was associated with a lower rate of maternal SARS-CoV-2

infection (1.0% vs 4.5%, aOR 0.17, 0.09–0.33, Table 3) and severe or critical COVID-19 (0.1% vs 0.6%, OR 0.15, 0.01–0.59). Due to the low frequency of maternal severe or critical COVID-19 in the vaccination group, multivariable modeling could not be performed. Prevalence of preeclampsia was similar between the two groups (8.1% vs 8.2%, aOR 0.82, 0.62–1.08).

Secondary analysis was performed investigating the primary outcome of perinatal death, restricting the cohort to only those patients with COVID-19 to determine whether disease severity was the primary driver of perinatal death. Only 20 of 2,069 vaccinated patients tested positive for SARS-CoV-2 infection,

Table 1. Demographics of the Study Population

Characteristic	Vaccinated (n=2,069)	Unvaccinated (n=13,796)	P	Missing Data	SMD	
					Prematch	Postmatch
Age (y)	31.56±5.15	27.83±5.8	<.01	16 (0.1)	0.681	−0.051
Nulliparous	904 (43.7)	4,963 (36.0)	<.01	7 (<0.1)	0.158	—*
Black race	517 (25.0)	5,276 (38.2)	<.01	254 (1.6)	0.289	—*
Hispanic ethnicity	163 (7.9)	1,368 (9.9)	<.01	278 (1.8)	0.073	—*
Twin gestation	37 (1.8)	213 (1.5)	.46	0 (0)	0.019	0.024
Public insurance	546 (26.4)	8,562 (62.1)	<.01	391 (2.5)	0.804	−0.043
BMI (kg/m ²)	27.94±7.37	29.32±8.15	<.01	10,836 (68.3)	0.177	0.007
Current tobacco use	52 (2.5)	1,349 (9.8)	<.01	237 (1.5)	0.310	−0.045
Diabetes	106 (5.1)	692 (5.0)	.87	3 (0.0)	0.004	−0.053
Hypertension	256 (12.4)	1,737 (12.6)	.79	19 (0.1)	0.006	0.059
HIV infection	3 (0.1)	37 (0.3)	.47	16 (0.1)	0.027	—*
Asthma	239 (11.6)	1,745 (12.6)	.16	11 (0.1)	0.033	—*

SMD, standardized mean difference; BMI, body mass index; HIV, human immunodeficiency virus.

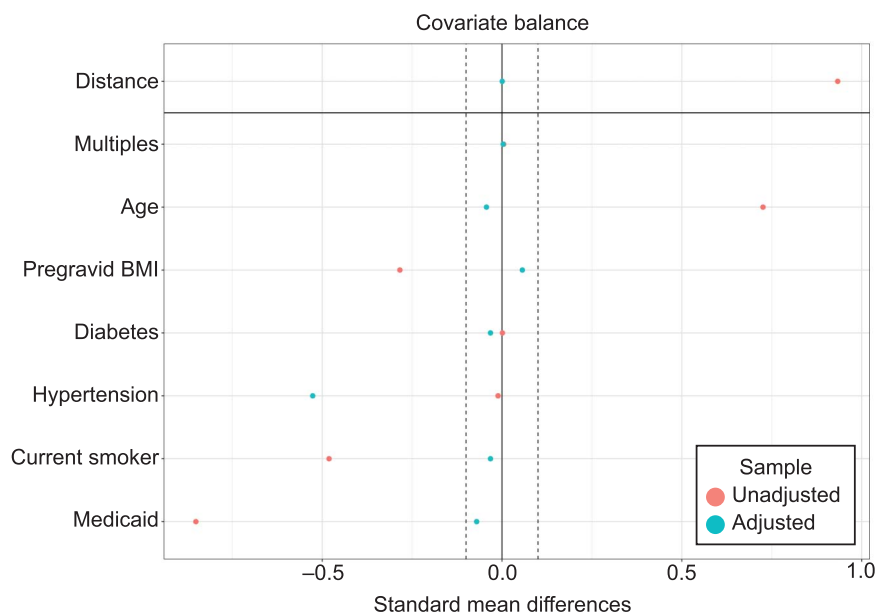
Data are mean±SD or n (%) unless otherwise specified.

* Variable not included in propensity score matching model.



Fig. 2. Love plot of propensity score using full matching with linear modeling based on propensity score estimate model, including adjusted and unadjusted covariate balance. BMI, body mass index.

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with no perinatal deaths; 5 of the 623 (0.8%) unvaccinated patients diagnosed with COVID-19 had the outcome of perinatal death. This difference was not statistically significant.

Propensity score matching was used to estimate the average marginal effect of vaccination on perinatal death, preterm delivery, very low birth weight, NICU admission, and critical or severe COVID-19. Full matching was performed due to the infrequency of outcomes. Among the 15,865 delivering patients, baseline BMI was missing for 10,836 (68.3%; 70% for unvaccinated patients and 54% for vaccinated patients). Given the large proportion of missing BMI data, propensity score matching was limited to patients with complete data, which included 938 vaccinated patients and 3,975 unvaccinated patients. The optimal propensity score estimate was generated using a generalized linear model linked by probit with full matching, including an interaction term for age and chronic hypertension. Match was balanced, with standardized mean difference less than 0.10 and a variance ratio for all variables between less than 2.0 and greater than 0.5 (Table 1, Fig. 2). Average treatment effect on the treated for vaccination did not remain a significant predictor of perinatal death (-0.0028 , $P=.248$, Table 2). Vaccination remained protective against other perinatal outcomes after propensity score matching, including preterm delivery (-0.0034 , $P=.007$), very low birth weight (-0.0104 , $P=.019$), and NICU admission (-0.0446 , $P=.001$). Vaccination likewise remained protective against critical or severe COVID-19 (average treatment effect on the treated -0.0085 , $P=.003$, Table 3).

DISCUSSION

In our large, retrospective cohort study, receipt of two doses of an mRNA COVID-19 vaccine was associated with improvements in both neonatal and maternal outcomes. Vaccinated patients had a 44% reduction in perinatal death, including a 38% reduction in stillbirth. Vaccination was also associated with lower rates of delivery before 37 weeks of gestation, indicated preterm birth, neonates with very low birth weight, and NICU admission. As previously reported, vaccination was associated with improved maternal outcomes including lower rates of SARS-CoV-2 infection and a reduction in severe or critical maternal COVID-19.⁴ When analyzed with propensity score matching, the protective effect of vaccination against perinatal death did not remain significant. However, the association with lower rates of preterm delivery, neonates with very low birth weight, NICU admission, and severe or critical maternal COVID-19 persisted.

Having COVID-19 during pregnancy has been associated with a range of adverse pregnancy outcomes, including twofold increased odds of stillbirth,⁸ and disease severity appears to modulate risks related to adverse outcomes.¹¹ However, the current literature on COVID-19 vaccination in pregnancy remains limited. Stock et al²⁴ published a large population-based study of 145,424 pregnant patients in Scotland, which found that vaccinated patients had a rate of perinatal mortality similar to the background population, whereas unvaccinated patients had an increased rate of perinatal mortality. A limitation of their study is that it is descriptive, because they were unable to adjust for confounders



Table 2. Perinatal Outcomes

Perinatal Outcome	Vaccinated (n=2,106)	Unvaccinated (n=14,026)	OR (95% CI)	aOR (95% CI)*	PSM	
					ATT [†]	P [‡]
Perinatal death [‡]	10 (0.5)	118 (0.8)	0.56 (0.29–1.07)	0.20 (0.05–0.88)	–0.0028	.247
Stillbirth	9 (0.4)	95 (0.7)	0.62 (0.31–1.24)	0.13 (0.02–0.99)		
Neonatal death	1 (0.05)	24 (0.17)	0.27 (0.01–1.70)	— [§]		
Preterm delivery	213 (10.3)	1,827 (13.2)	0.75 (0.64–0.87)	0.63 (0.48–0.82)	–0.0034	.007
Spontaneous	92 (4.4)	858 (6.2)	0.70 (0.56–0.87)	0.71 (0.48–1.03)		
Indicated	116 (5.6)	945 (6.8)	0.80 (0.66–0.98)	0.60 (0.42–0.86)		
SGA [¶]	86 (4.1)	654 (4.7)	0.79 (0.63–0.99)	1.15 (0.79–1.68)		
Very low birth weight [#]	23 (1.1)	294 (2.1)	0.51 (0.33–0.79)	0.35 (0.15–0.84)	–0.0104	.019
NICU admission	249 (11.8)	2,141 (15.3)	0.74 (0.64–0.85)	0.66 (0.52–0.85)	–0.0446	.001

OR, odds ratio; aOR, adjusted odds ratio; PSM, propensity score matching; ATT, estimated average treatment effect on the treated; SGA, small for gestational age; NICU, neonatal intensive care unit.

Data are n (%).

* Logistic regression analysis was used to calculate an aOR for the primary outcome to control for age, prepregnancy body mass index, diabetes, hypertension, current smoking, public insurance, and multiple gestations.

[†] Generated by propensity score matching, including only patients with completed variables (n=4,913).

[‡] A combination of stillbirth (nonanomalous fetal death after 20 weeks of gestation) and neonatal death (death between birth and 28 days of life).

[§] Insufficient counts to allow multivariable analysis for neonatal death.

^{||} Delivery before 37 weeks of gestation. n=15,865 for preterm delivery, spontaneous preterm delivery, and indicated preterm delivery, reflecting 15,865 deliveries of 16,132 neonates (multiples).

[¶] Birth weight less than the 10th percentile on the Fenton curve¹⁴ by neonatal sex.

[#] Birth weight less than 1,500 g.

due to data-set limitations.²⁴ Remaining available literature is mostly limited to reporting vaccine safety with no difference in adverse neonatal outcomes between vaccinated and unvaccinated pregnant patients.^{5,13–15} These studies are limited by sample size, inclusion of partially vaccinated patients, or both.

In adjusted analysis, we found that patients not previously infected with SARS-CoV-2 who received the mRNA COVID-19 vaccine prior to delivery had a decreased risk of both perinatal death and stillbirth. The low overall frequency of perinatal death results in an

unstable point estimate with significance margin approaching unity. Although the protective association against perinatal death persisted in a sensitivity analysis of only singleton gestations, the estimated average treatment effect on the treated generated by propensity score matching did not achieve significance. However, the summation of our findings reiterates that COVID-19 vaccination is protective against a range of adverse pregnancy outcomes.

Strengths of the current study include a large, diverse study population from a multistate hospital

Table 3. Maternal Outcomes

Maternal Outcome	Vaccinated (n=2,069)	Unvaccinated (n=13,796)	OR (95% CI)	aOR (95% CI)*	Completed Match	
					ATT [†]	P [‡]
SARS-CoV-2 infection	20 (1.0)	616 (4.5)	0.21 (0.13–0.32)	0.17 (0.09–0.33)		
At delivery admission	8 (0.4)	164 (1.2)	0.32 (0.15–0.65)	0.48 (0.17–1.41)		
Severe or critical COVID-19 [‡]	2 (0.1)	83 (0.6)	0.15 (0.01–0.59)	— [§]	–0.0085	.003
Preeclampsia	167 (8.1)	1,127 (8.2)	0.98 (0.83–1.16)	0.82 (0.62–1.08)		

OR, odds ratio; aOR, adjusted odds ratio; ATT, estimated average treatment effect on the treated; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2; COVID-19, coronavirus disease 2019.

* Logistic regression analysis was used to calculate an aOR for the primary outcome to control for age, prepregnancy body mass index, diabetes, hypertension, current smoking, public insurance, and multiple gestations.

[†] Generated by propensity score matching including only patients with completed variables (n=4,913).

[‡] According to the National Institutes of Health definition.¹⁷

[§] Due to low occurrence of maternal severe or critical COVID-19 in the vaccinated group, our logistic regression model was unable to calculate an aOR for this outcome.



system with a variety of care sites encompassing academic, urban community, and rural medical practices, which allows for generalizability. Strict inclusion criteria allowed our study to focus on patients who completed a two-dose mRNA COVID-19 vaccine series, excluding partially vaccinated patients and those with viral vector vaccines. Inclusion of all patients who received two doses of the vaccine before delivery, not only those vaccinated during the index pregnancy, allowed us to evaluate the effect of the COVID-19 vaccine administration any time before delivery. The findings add to the growing literature supporting efficacy of COVID-19 vaccination in pregnant individuals.

Our study has several limitations, particularly its retrospective nature. Though data on vaccination status, stillbirth, and preterm delivery were validated with manual medical record review, bias may be introduced through errors in data entry by medical practitioners. A large proportion of our patient population did not have a prepregnancy or first-trimester BMI available, because many practitioners who deliver patients at an Ochsner Health Facility provide prenatal care at external sites with separate medical record systems, either electronic or physical, that were unavailable for review. We cannot account for patients vaccinated outside of Ochsner Health at a site that did not upload its vaccination data into the LINKS system, though this would be more likely to introduce type II error by misclassifying vaccinated patients as unexposed. The vaccination rate improved throughout the study period consistent with national trends of increasing vaccine acceptance in pregnancy, as well as increasing vaccine availability. Unvaccinated patients were shown to have a higher rate of adverse maternal and neonatal outcomes. This group was also found to have a higher proportion of patients of Black racial and Hispanic ethnic backgrounds, and those with Medicaid insurance. Race was not used in our logistic regression analysis because it is a social construct and not a biological determinant of adverse pregnancy outcomes. Due to the timing of the study period, we are unable to draw conclusions regarding the protective effect of “booster” doses (third or fourth doses) of the mRNA vaccines, which are now being offered to the general population, including pregnant and lactating patients. Finally, as the virus continues to mutate, we are unable to speak to the protection offered by vaccination in potentially less pathogenic waves,¹⁶ nor to the benefits of next-generation formulations of the vaccines.

In our large retrospective cohort study, receipt of the primary mRNA COVID-19 vaccination series

was associated with a lower rate of adverse pregnancy outcomes, including perinatal death, preterm delivery, neonates with very low birth weight, and NICU admission. Although the decreased risk of perinatal death did not persist in propensity score matching analysis, there was evidence of directional benefit for vaccinated patients. Our findings support the continued strong recommendation by the CDC, the American College of Obstetricians and Gynecologists, and the Society for Maternal-Fetal Medicine for vaccination of all pregnant patients against COVID-19.¹⁻³

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